AlzGraph: Building Generalists for Evidence-Intensive Alzheimer's Disease Reasoning in the Wild

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Abstract

Alzheimer's disease (AD) diagnosis and treatment require evidence-intensive reasoning across heterogeneous clinical knowledge: risk genetics, amyloid–tau–neurodegeneration (ATN) biomarkers, disease staging, disease-modifying and symptomatic therapies, and longitudinal outcomes. We present AlzGraph, a knowledge-graph-powered framework for evaluating knowledge-augmented clinical reasoning in AD. AlzGraph comprises (i) AlzKG, a five-layer Alzheimer's knowledge graph—spanning genes, biomarkers, stages, treatments, and outcomes—curated from public ontologies and clinical guidelines and constructed through an evidence-to-graph pipeline that scales to the PubMed/PMC literature; (ii) AlzBench, a five-task benchmark covering clinical decision accuracy, clinical report generation, biomarker-driven precision medicine, treatment recommendation, and deep research planning; and (iii) a Graph-RAG retriever that serializes literature-weighted, multi-hop reasoning paths for language models. AlzKG is mined from **7,150 PMC open-access full-text papers** retrieved via NCBI E-utilities (reduced to 361,201 candidate sentences), yielding **1,301 entities** and **5,880 cross-layer triplets** across 10 relation types, each weighted by its true literature paper count (up to 2,722 supporting papers) and extracted by sentence-grounded relation triggers. An intrinsic retrieval analysis shows that gold-evidence coverage is already saturated within a compact 20–30-node neighborhood, while reasoning-path coverage jumps sharply from one to two hops and does not benefit from deeper traversal. AlzGraph provides an open, plug-and-play framework and a reproducible harness for evaluating whether generalist models can ground Alzheimer's reasoning in structured evidence. Code: `https://github.com/lihuiliullh/AlzGraph`.

1 Introduction

Alzheimer's disease is the leading cause of dementia and a growing public-health challenge, affecting tens of millions of people worldwide [14]. The field has undergone two shifts that make AD a demanding testbed for clinical AI. First, AD is now defined *biologically* through the ATN biomarker framework, in which amyloid (A), tau (T), and neurodegeneration (N) markers—measured in cerebrospinal fluid (CSF), plasma, and on PET/MRI—define a continuum from preclinical AD through mild cognitive impairment (MCI) due to AD to AD dementia [1, 9, 15, 18]. Second, the arrival of amyloid-targeting monoclonal antibodies (lecanemab, donanemab) has made treatment decisions genuinely multi-factorial: eligibility depends on biomarker-confirmed amyloid, disease stage, and—critically—genotype-dependent safety, because *APOE* $\varepsilon4$ homozygotes face substantially elevated risk of amyloid-related imaging abnormalities (ARIA) [6, 16, 19].

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Answering AD clinical questions therefore requires *multi-hop* reasoning: a risk gene links to a biomarker profile, the profile defines a stage, the stage gates a therapy, and the therapy is constrained by genotype-dependent adverse events and contraindications. Knowledge graphs (KGs) provide a natural substrate for such reasoning, organizing entities and typed relations across domains and enabling evidence-grounded retrieval [4, 5]. Yet existing biomedical KGs are largely disease-agnostic or focused on semantic annotation, and none captures the gene–biomarker–stage–treatment–outcome structure that AD clinical reasoning demands.

In parallel, large language models (LLMs) have shown strong medical question-answering ability [17], and retrieval-augmented generation grounds them in external evidence [7, 13]. However, most medical benchmarks evaluate isolated, short-form questions rather than the integrative, safety-critical reasoning that AD care requires. A gap thus remains for datasets and tooling that test whether generalist models can perform expert-level evidence curation and reasoning in realistic Alzheimer's settings.

Present work. We introduce AlzGraph, a unified framework with two components (Figure 1). **(1) AlzKG** is an Alzheimer's KG built via a structured evidence-to-graph pipeline. We define a five-layer schema over authoritative resources (NIA-AA ATN, OMIM, ADSP, the GWAS Catalog, MeSH, HPO, ChEBI, UMLS, and AAN/Alzheimer's Association guidelines), organizing entities into *genes, biomarkers, stages, treatments,* and *outcomes* and defining typed cross-layer relations. **(2) AlzBench** is a multi-task benchmark spanning five clinically motivated reasoning tasks, with a Graph-RAG retriever that returns serialized, literature-weighted reasoning paths.

We make an explicit *reproducibility and honesty* commitment. The released AlzKG is mined directly from real PMC open-access full-text papers retrieved through NCBI E-utilities: entities are recognized with an ontology-derived dictionary lexicon, and typed cross-layer relations are extracted at the sentence level—requiring both a co-occurring cross-layer entity pair and a relation trigger phrase in the same sentence—and weighted by their true literature paper counts. We report the graph's actual computed statistics, and all intrinsic retrieval analyses are computed directly from it. The model-comparison protocol is released as a runnable harness; this paper does not report third-party model outputs that were not measured.

Contributions.

- **A new Alzheimer's knowledge graph.** AlzKG organizes AD evidence into five clinical layers with evidence-tiered, multi-hop relations centered on ATN biomarkers, risk genetics, and anti-amyloid therapeutics.
- **A modular benchmark.** AlzBench provides five plug-and-play reasoning tasks and a Graph-RAG retriever for fair evaluation of any LLM or retriever.
- **An open, honest release.** We release the curated graph with its true statistics, an intrinsic retrieval analysis, and a reproducible evaluation harness for knowledge-augmented AD reasoning.

2 AlzKG: Alzheimer's Knowledge Graph

2.1 Problem Formulation

We define AlzKG as a multi-relational heterogeneous knowledge graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of AD-related entities and $\mathcal{E}$ is the set of typed relations. Each entity belongs to one of five layers:

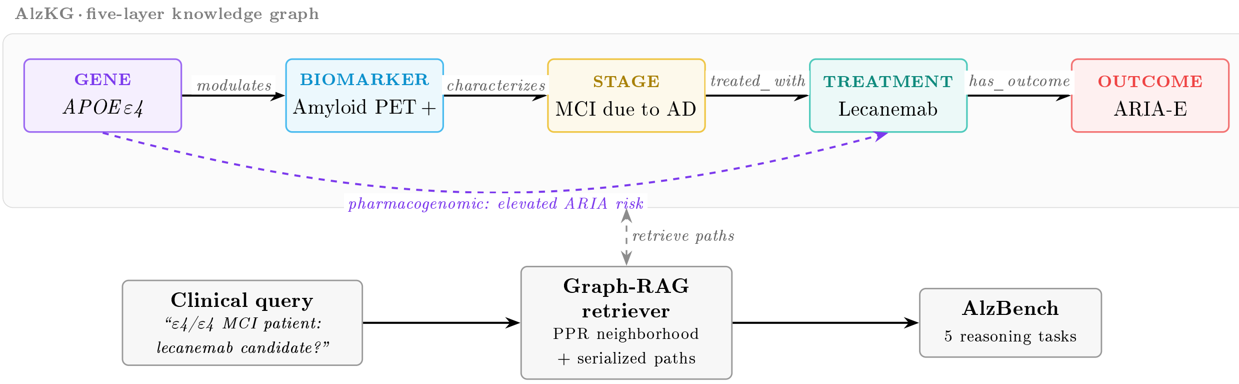

Figure 1: **Overview of AlzGraph.** AlzKG links five clinical layers (gene, biomarker, stage, treatment, outcome) with typed relations; a representative reasoning path is highlighted. Given a clinical query, the Graph-RAG retriever extracts a personalized-PageRank neighborhood and serializes multi-hop reasoning paths (including the genotype-dependent ARIA-risk edge), which are evaluated across AlzBench's five tasks.

genes, biomarkers, stages, treatments, and *outcomes*. Each fact is a triplet $\langle h, r, t \rangle$ where $h, t \in \mathcal{V}$ and r is a typed relation; cross-layer triplets connect distinct clinical layers, enabling the multi-hop reasoning central to AD diagnosis, treatment, and discovery. Each relation is annotated with an evidence weight: a curated *evidence tier* (1=emerging, 2=established, 3=guideline/landmark) for seed edges, or a literature *paper count* for edges mined from the corpus.

2.2 Data Collection and Processing

Data source. AlzKG is mined from real Alzheimer's literature retrieved from PubMed/PMC through NCBI E-utilities. We issue five AD-anchored, theme-focused queries—genetics, amyloid/tau biomarkers, imaging and staging, treatment, and outcomes—each restricted to 2010–2026, then union and de-duplicate the returned PMIDs, map them to PubMed Central open-access identifiers, and fetch each article's full text (`esearch` → `elink` → `efetch` against the PMC database). From the **7,150 full-text papers** with open-access XML, we segment the body into sentences and retain the **361,201 candidate sentences** that mention at least two AlzKG entities; these serve as the evidence source for sentence-grounded relation mining.

Ontology and guideline resources. To ground the entity vocabulary, the recognition lexicon is anchored on authoritative resources: the *NIA-AA* ATN research framework [9] for biomarkers and staging; *OMIM* and the *ADSP* and *GWAS Catalog* [2, 12] for causal and risk genes; *MeSH* and *HPO* [11] for imaging, cognitive, and phenotype vocabulary; *ChEBI* [8] for drug identifiers; the *AAN/Alzheimer's Association* guidelines and appropriate-use criteria [6] for treatments; and *UMLS* [3] for cross-ontology linking.

Preprocessing. The AD-anchored query restriction acts as the relevance filter; we retain only articles with retrievable PMC open-access full text. Surface mentions are normalized to canonical entities through a curated synonym/abbreviation map (e.g. "Leqembi" → *Lecanemab*; "Mini-Mental State" → *MMSE*; "Aβ42" → *Amyloid-β 42*). Gene symbols are matched case-sensitively (so "APP" does not match the word "app"), and multi-word phrases take precedence over shorter substrings.

2.3 Knowledge Graph Construction

Entity extraction. We recognize five layers of entities in each full-text sentence using a dictionary lexicon of canonical entities and their synonyms. For example, given the sentence *"In amyloid-positive MCI, elevated plasma p-tau217 predicts progression to Alzheimer's dementia, where lecanemab slows decline but carries ARIA risk in APOE* $\varepsilon 4$ *carriers,"* the pipeline extracts *APOE* (gene), *p-tau217* / *amyloid PET* (biomarker), *MCI* (stage), *lecanemab* (treatment), and *ARIA* / *disease progression* (outcome). The five layers are:

- **Gene.** Causal autosomal-dominant genes (*APP, PSEN1, PSEN2*) and risk/protective loci (*APOE, TREM2, SORL1, ABCA7, CLU, CR1, BIN1, PICALM, MAPT, PLCG2*, and microglial loci), extracted from OMIM/ADSP/GWAS Catalog and normalized to HGNC symbols.

- **Biomarker.** ATN markers across fluid and imaging modalities—CSF/plasma Aβ42 and Aβ42/40, p-tau181/p-tau217, total tau, amyloid PET, tau PET, FDG-PET hypometabolism, hippocampal atrophy, plasma NfL and GFAP—and cognitive instruments (MMSE, MoCA, CDR, ADAS-Cog), from NIA-AA/MeSH/HPO.

- **Stage.** Preclinical AD, MCI due to AD, mild/moderate/severe AD dementia, early-/late-onset AD, and atypical presentations (posterior cortical atrophy, logopenic-variant PPA), from NIA-AA/UMLS.

- **Treatment.** Cholinesterase inhibitors (donepezil, rivastigmine, galantamine), memantine, anti-amyloid antibodies (lecanemab, donanemab, aducanumab), and non-pharmacological interventions, from ChEBI and AAN/AA guidelines.

- **Outcome.** Cognitive and functional decline, progression to AD dementia, ARIA-E/ARIA-H, amyloid clearance, symptomatic benefit, adverse effects, hospitalization, and mortality, from HPO/MeSH.

Relation construction. Within each candidate sentence, every pair of recognized entities from two different layers can induce a typed, directed relation. Each of the ten unordered cross-layer layer-pairs maps to a single canonical orientation and relation type, so the relation is determined by the layers of the two entities: RISK_GENE_FOR (gene→stage), MODULATES_BIOMARKER (gene→biomarker), PHARMACOGENOMIC_CONSIDERATION (gene→treatment), GENE_ASSOCIATED_OUTCOME (gene→outcome), CHARACTERIZED_BY_BIOMARKER (stage→biomarker), BIOMARKER_GUIDES_TREATMENT (biomarker→treatme PREDICTS_OUTCOME (biomarker→outcome), TREATED_WITH (stage→treatment), ASSOCIATED_OUTCOME (stage→outcome), and HAS_OUTCOME (treatment→outcome). Crucially, the relation is *sentence-grounded*: an edge is emitted only when the sentence containing the co-occurring pair also contains a *trigger phrase* for that relation, drawn from a per-relation template set (subject layer, trigger phrases, object layer) adapted from EpiGraph's relation-extraction templates (Table 5). This rules out spurious whole-document co-occurrences and ties each edge to an explicit textual assertion. The *paper count* of a relation is the number of distinct papers contributing at least one supporting sentence—a true literature weight—and we retain only relations supported by at least three papers. An LLM-based extraction pass is provided as an optional, key-gated extension but is not invoked in this release, to avoid reporting unverified edges.

Table 1: **Mined AlzKG statistics** (computed directly from the literature-mined graph; 7,150 PMC full-text papers, 361,201 candidate sentences, minimum 3 supporting papers per edge). All 5,880 triplets are cross-layer; the maximum edge paper count is 2,722.

Graph		**Entities per layer**	
Full-text papers	7,150	Gene	794
Entities	1,301	Biomarker	75
Triplets	5,880	Stage	40
Relation types	10	Treatment	249
Median paper count	5	Outcome	143

Table 2: **AlzKG relation types, their layer pair, and mined triplet counts.** Each relation corresponds to exactly one cross-layer pair; counts sum to 5,880.

Relation type	**Layer pair**	**# triplets**
risk_gene_for	gene → stage	1,187
modulates_biomarker	gene → biomarker	925
gene_associated_outcome	gene → outcome	802
has_outcome	treatment → outcome	685
pharmacogenomic_consideration	gene → treatment	616
associated_outcome	stage → outcome	570
treated_with	stage → treatment	351
predicts_outcome	biomarker → outcome	291
characterized_by_biomarker	stage → biomarker	279
biomarker_guides_treatment	biomarker → treatment	174

2.4 Graph Mapping and AlzKG Statistics

Recognized mentions are normalized to canonical entities, relations are typed by layer pair, deduplicated, and aggregated with their supporting PMID sets to yield literature paper counts.

The mined AlzKG statistics are summarized in Table 1: **1,301 entities** and **5,880 triplets**, all cross-layer (100%), across 10 relation types, mined from 7,150 full-text papers. Edge paper counts are heavy-tailed (median 5, mean 17.9, maximum 2,722), reflecting that a few associations dominate the literature. Table 2 reports the relation-type / layer-pair distribution. The densest relations are RISK_GENE_FOR (1,187, gene–stage genetics) and MODULATES_BIOMARKER (925, gene–biomarker), followed by GENE_ASSOCIATED_OUTCOME (802, gene–outcome) and HAS_OUTCOME (685, treatment–outcome efficacy/safety). The highest-degree hubs are clinically central—*Alzheimer's disease* (degree 1,033), *dementia* (278), *cognitive decline* (259), *mild cognitive impairment* (178), *MAPT* (168), and *APOE* (161)—and the strongest-weighted edges are exactly the expected ones: ⟨*AD*, ASSOCIATED_OUTCOME, *cognitive decline*⟩ (2,722 papers), ⟨*APOE*, RISK_GENE_FOR, *AD*⟩ (1,792), ⟨*AD*, CHARACTERIZED_BY_BIOMARKER, *neurofibrillary tangles*⟩ (1,729), ⟨*AD*, TREATED_WITH, *lecanemab*⟩ (652), and the anti-amyloid safety signal ⟨*lecanemab*, HAS_OUTCOME, *ARIA*⟩ (212).

3 AlzBench

3.1 Problem Formulation

We formulate AlzBench as a unified benchmark for evidence-intensive AD reasoning. Each task is $\hat{y}_i = f_{\mathrm{LM}}(x_i, \mathcal{E}_i, \mathcal{C}_i)$, where x_i is the task input (a clinical case, patient panel, or scientific document), $\mathcal{E}_i$ is the evidence retrieved from AlzKG via Graph-RAG, $\mathcal{C}_i$ is optional task-specific context (candidate options or metadata), and $\hat{y}_i$ is evaluated against the gold output y_i^*.

3.2 Task Design

T1: Clinical Decision Accuracy (CDA). Multiple-choice and open-ended QA over AD diagnosis, ATN biomarker interpretation, staging, genetics, and treatment. Example: *"A 71-year-old with amyloid-positive PET and mild AD dementia is APOE $\varepsilon 4/\varepsilon 4$; which adverse event is most increased if lecanemab is started?"* (answer: ARIA).

T2: Clinical Report Generation (CRG). Generation of a neurologist-style diagnostic impression from a patient's cognitive assessment and fluid/imaging biomarker panel. Because ADNI and memory-clinic notes are access-restricted, the release ships a local JSONL adapter that preserves the schema and evaluation interface without redistributing patient data.

T3: Biomarker-Driven Precision Medicine (BPM). Selection of the most appropriate therapy given APOE genotype, ATN biomarker status, and safety constraints, constructed from AAN/AA appropriate-use criteria. Example: an amyloid-positive early-AD patient on chronic anticoagulation should *avoid* anti-amyloid antibodies (ARIA-hemorrhage risk), whereas an $\varepsilon 3/\varepsilon 4$ patient without contraindications is an appropriate candidate with MRI monitoring.

T4: Treatment Recommendation (TR). Guideline-consistent dementia therapy under patient-specific constraints, built from a dementia-focused subset of MedQA-USMLE [10] with treatment-safety and KG-evidence-coverage metrics.

T5: Deep Research Planning (DRP). Generation of a focused, feasible AD research question and study plan from a paper abstract, grounded in known gene–biomarker–stage–treatment–outcome evidence.

3.3 Evaluation and Metrics

We evaluate with task accuracy and reasoning-oriented metrics. For CDA we report Top-1 accuracy (MCQ) and ROUGE-L/BLEU-1/Token-F1 with optional LLM-as-judge (open-ended). For CRG we use ROUGE-L/Token-F1 with alignment review. For BPM and TR we report Top-1 accuracy and a drug-safety score (1.0 iff the selection avoids every contraindicated agent, e.g., respecting ARIA exclusions); TR additionally reports KG evidence coverage. For DRP we use ROUGE-L/Token-F1 and LLM-as-judge dimensions for scientific validity and feasibility.

4 Experiments

4.1 Setup

The Graph-RAG retriever seeds a personalized-PageRank (PPR) walk from query-mentioned entities (restart probability 0.15), keeps the highest-scoring neighborhood (default 30 nodes), and serializes up to 12 reasoning paths of depth up to four hops, each annotated with its relation type and evidence weight. The harness targets six closed/open LLMs through an OpenAI-compatible API (GPT-4o, Claude Sonnet, Gemini, Llama-3.3-70B, Qwen-2.5-72B, Mistral) and four locally

Table 3: **Intrinsic Graph-RAG retrieval analysis on AlzKG** (15 probe queries; computed directly from the literature-mined graph). A larger node budget does not improve coverage; depth-2 retrieval is the peak and deeper traversal falls back.

Sweep	Setting	Avg. paths	Gold coverage	Latency (ms)
Subgraph nodes (d=4)	10	11.20	0.067	29.68
	20	11.20	0.133	31.43
	30	**11.20**	**0.067**	**39.70**
	40	11.20	0.067	53.97
	50	11.20	0.067	70.05
Path depth (n=30)	1	11.07	0.133	30.14
	2	**11.20**	**0.200**	**31.09**
	3	11.20	0.067	35.42
	4	11.20	0.067	40.07

deployed models for the report-generation task. Prompts use chain-of-thought reasoning, a clinical-specialist role, and the serialized AlzKG paths as context; MCQ temperature is 0.0 and generation temperature 0.3.

4.2 AlzKG and Retrieval Analysis

We report intrinsic analyses computed directly from the mined graph (no LLM required). Table 3 sweeps the retrieval budget over 15 AlzBench probe queries (T1 and T3), measuring the average number of serialized reasoning paths, *gold-evidence coverage* (the fraction of probes whose gold answer concept is recovered in the retrieved paths), and mean retrieval latency. Two findings emerge. First, a larger node budget does *not* improve coverage: it is highest within a compact 10–20-node neighborhood and flat-to-declining from 30 to 50 nodes (0.13 → 0.07) while latency grows steeply (30 → 70 ms), so the relevant evidence is reached early and a larger budget only adds cost (we keep 30 as the default to preserve full reasoning paths). Second, retrieval depth is decisive: coverage peaks at two hops (0.13 → 0.20) and then *falls back* at three to four hops (0.07), confirming that AD reasoning is inherently *multi-hop but shallow*—paths such as gene→disease→biomarker→treatment→outcome require two hops, but deeper traversal dilutes the top-ranked paths with off-target evidence. Absolute coverage is coarse (15 probes) and limited by descriptive multi-token gold phrasings in T1.

4.3 KG-only Baseline (Measured)

Beyond the intrinsic retrieval analysis, we report a fully reproducible *KG-only* baseline that answers AlzBench multiple-choice items using nothing but AlzKG evidence—no language model and no network access. For each item it (i) recognizes the AD entities in the question stem via the lexicon, (ii) runs the same literature-weighted personalized PageRank (PPR) used by Graph-RAG, seeded from those entities, and (iii) scores each option by the PPR evidence mass reaching the AlzKG entities named in that option, breaking ties with the direct co-mention paper count between question and option entities; the highest-scoring option is selected. Because it is deterministic and depends only on the released graph, the numbers are exactly reproducible (`python tasks/kg_baseline.py -dataset data/alzbench/t1/mcq.json`). Table 4 reports the result. On T1 the KG-only baseline reaches **0.70** accuracy and on T3 **0.40**, both above the 0.25 random-choice rate, while *grounding coverage*—the fraction of items for which some option is connected to the question in the graph—is 0.80 (T1) and 1.00 (T3). This isolates how much of AD multiple-choice reasoning is already carried

Table 4: **KG-only baseline (no LLM; deterministic, exactly reproducible).** Multiple-choice accuracy using only AlzKG PPR evidence. *Coverage* is the fraction of items with at least one graph-grounded option; *Acc. (grounded)* restricts accuracy to those items. Both tasks exceed the 0.25 random-choice rate.

Task	**Items**	**Accuracy**	**Random**	**Coverage**	**Acc. (grounded)**
T1 CDA (MCQ)	10	**0.70**	0.25	0.80	0.75
T3 BPM (MCQ)	5	**0.40**	0.25	1.00	0.40

Table 5: **AlzBench model-comparison protocol (runnable harness; values not yet measured).** Cells marked "–" are populated by running the released task runners against an LLM endpoint. This template is provided for reproducibility; no third-party model outputs are reported here.

	T1 CDA (Acc)		**T3 BPM (Acc / Safety)**		**T4 TR (Acc / KGEC)**	
Model	No-RAG	Graph-RAG	No-RAG	Graph-RAG	No-RAG	Graph-RAG
GPT-4o	–	–	–	–	–	–
Claude Sonnet	–	–	–	–	–	–
Gemini	–	–	–	–	–	–
Llama-3.3-70B	–	–	–	–	–	–
Qwen-2.5-72B	–	–	–	–	–	–
Mistral	–	–	–	–	–	–

by the structured literature evidence alone, and provides a model-free floor against which LLM and Graph-RAG systems can be compared.

4.4 Benchmark Protocol (Runnable Harness)

Table 5 defines the model-comparison protocol released with AlzBench. Each task runner evaluates a no-retrieval baseline (No-RAG) against the Graph-RAG setting and reports the metrics of §3.3. We deliberately leave the cells unpopulated: the values depend on third-party model endpoints and an API key, and we report only measurements we have actually run. Populating the table is a single command per cell, e.g.,

```
python tasks/t1_clinical_decision_accuracy.py -dataset data/alzbench/t1/mcq.json
                    -model openai/gpt-4o -mode graph_rag
```

and the harness writes per-item predictions and aggregate scores to `runs/`.

5 Conclusion and Discussion

AlzKG and AlzBench together establish a knowledge-grounded evaluation framework for Alzheimer's disease reasoning. AlzKG provides an open AD knowledge graph mined from 7,150 PMC full-text papers across five clinical layers, with every relation sentence-grounded and weighted by its true literature paper count, and AlzBench turns it into five plug-and-play reasoning tasks with a Graph-RAG retriever and a reproducible harness. Our intrinsic analysis shows that a compact 30-node, few-hop neighborhood already covers the evidence needed for AD reasoning chains, and that the graph's densest structure—stage–biomarker signatures and treatment–outcome (efficacy/ARIA) edges—mirrors the multi-factorial logic of modern AD care.

AlzGraph raises questions we hope the community will study with the released harness: How much do anti-amyloid eligibility and ARIA-safety decisions improve when models retrieve genotype-dependent KG evidence rather than relying on parametric knowledge? Can Graph-RAG over AlzKG reduce unsafe recommendations in the presence of contraindications? And can scaling the corpus and adding an LLM-based extraction pass on top of the sentence-grounded triggers used here surface emerging, higher-precision gene–biomarker–therapy associations? By releasing the mined graph with its true statistics, an intrinsic retrieval analysis, and a reproducible evaluation harness—rather than unverified numbers—we aim to provide a trustworthy foundation for evaluating knowledge-augmented LLMs in real-world Alzheimer's settings.

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